E-CERTIFICATE VALIDATION AND VERIFICATION USING BLOCKCHAIN

PROJECT REPORT

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to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the Degree

of

Bachelor of Technology

In

Computer Science and Engineering



Department of Computer Science and Engineering

Rajadhani Institute of Engineering and Technology

Nagaroor

May 2019

DECLARATION

We undersigned hereby declare that the project report ***'E-Certificate Validation And Verification Using Blockchain'*****,** submitted for partial fulfilment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of ***Ms. VISHAGINI V***. This submission represents our idea in our own words and where ideas or words of others have been included, We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Nagaroor
Date: 12/06/2019

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CERTIFICATE

This is to certify that the report entitled '**E-CERTIFICATE VALIDATION AND VERIFICATION USING BLOCKCHAIN** ' submitted by **BLESSON P KOSHY,** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by him under our guidance and supervision.. This report in any form has not been submitted to any other University or Institute for any purpose.

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ACKNOWLEDGEMENT

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We are extremely happy to mention a great word of gratitude to **Ms. Sangeetha Shibu,** Associate Professor, Head of the Department of Computer Science and Engineering, for providing us with all facilities for the completion of this work.

We would like to express our sincere gratitude to our project guide **Ms. Vishagini V**, Assistant Professor, Dept. of Computer Science and Engineering for her relevant suggestions, guidance and inestimable leadership given in completing this project report.

Finally, yet importantly, we would also like to express our gratitude to the project coordinator, **Ms. Beena K R,** Assistant Professors, Dept. of Computer Science and Engineering, for her valuable assistance provided during the course of the project. We would also extend our gratefulness to the rest of the faculties of our department. We also thank all our friends and well-wishers who greatly helped us in our endeavor.

APARNA B

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ABSTRACT

According to the Taiwan Ministry of Education statistics, about one million graduates each year, some of them will go to countries, high schools or tertiary institutions to continue to attend, and some will be ready to enter the workplace employment. During the course of study, the students' all kinds of excellent performance certificates, score transcripts, diplomas, etc., will become an important reference for admitting new schools or new works. As schools/college make various awards or diplomas, only the names of the schools and the students are input.

Due to the lack of effective anti-forge mechanism, events that cause the graduation certificate to be forged often get noticed. In order to solve the problem of counterfeiting certificates, the digital certificate system based on Blockchain technology would be proposed. By the unmodifiable property of Blockchain, the digital certificate with anti-counterfeit and verifiability could be made. The procedure of issuing the digital certificate in this system is as follows. First, generate the electronic file of a paper certificate accompanying other related data into the database, meanwhile calculate the electronic file for its hash value. Finally, store the hash value into the block in the chain system. The system will create a related QR-code and inquiry string code to affix to the paper certificate. It will provide the demand unit to verify the authenticity of the paper certificate through mobile phone scanning or website inquiries. Through the unmodifiable properties of the Blockchain, the system not only enhances the credibility of various paper-based certificates, but also electronically reduces the loss risks of various types of certificates

CONTENT

Content	Page No.
ACKNOWLEDGEMENT	**i**
ABSTRACT	**ii**
LIST OF FIGURES	**v**
LIST OF ABBREVIATIONS	**vi**
Chapter **1.** INTRODUCTION	**1**
1.1. General Background	**2**
1.2. Objectives	**3**
1.3. System Specifications	
Chapter **2.** LITERATURE SURVEY	**4**
2.1. Blockchain	**4**
2.2. Ethereum	**5**
2.3. Ethereum Virtual Machine	**6**
2.4. Smart Contract	**7**
2.5. Solidity	**8**
Chapter **3.** RESEARCH METHOD	**9**
3.1. System Design	**9**
3.2. Process	**10**
Chapter **4.** PROPOSED SYSTEM	**11**
4.1. Module Description	**11**
4.2. System Features	**11**
4.3. Modules	**12**

Chapter **5.** DESIGN **13**

5.1. UML Diagrams **13**

5.2. Sequence Diagram **15**

Chapter **6.** RESULTS **16**

Chapter **7.** SYSTEM STUDY **24**

7.1. Feasibility Study **24**

7.2. Non Functional Requirements **25**

7.3. Accuracy **25**

7.4. Usability **25**

7.5. Accessibility **25**

7.6. Performance **25**

7.7. Reliability **26**

7.8. Security **26**

Chapter **8.** EVALUATION **27**

Chapter **9.** CONCLUSION **30**

REFERENCES **31**

LIST OF FIGURES

No	Figure	Page No.
1	Blockchain Technology	2
2	Blockchain Technology version	4
3	EVM	7
4	Smart Contract	8
5	System Design	9
6	Process	10

LIST OF ABBREVIATIONS

EVM	Ethereum Virtual Machine
LLL	Low-Level Language
UML	Unified Modeling Language
PHP	Hypertext Preprocessor
HTML	Hypertext Markup Language
CSS	Cascading Style Sheet
IDE	Integrated Development Environment

CHAPTER 1
INTRODUCTION

1.1 General Background

Advances in information technology, the wide availability of the Internet, and common usage of mobile devices have changed the lifestyle of human beings. Virtual currency, digital coins originally designed for use online, has begun to be extensively adopted in real life. Because of the convenience of the Internet, various virtual currencies are thriving, including the most popular Bitcoin, Ether, and Ripple the value of which has surged recently. People are beginning to pay attention to blockchain, the backbone technology of these revolutionary currencies. Blockchain features a decentralized and incorruptible database that has high potential for a diverse range of uses. Blockchain is a distributed database that is widely used for recording distinct transactions.

Once a consensus is reached among different nodes, the transaction is added to a block that already holds records of several transactions. Each block contains the hash value of its last counterpart for connection. All the blocks are connected and together they form a blockchain . Data are distributed among various nodes (the distributed data storage) and are aw121thus decentralized. Consequently, the nodes maintain the database together. Under blockchain, a block becomes validated only once it has been verified by multiple parties. Furthermore, the data in blocks cannot be modified arbitrarily. A blockchain-based smart contract, for example, creates a reliable system because it dispels doubts about information's veracity.

Because information technology has developed rapidly in recent years, data protection is more necessary than ever. Graduates, whether they choose to continue studying or start job hunting, require various certificates for interviews. However, they often find that they have lost their educational and commendation certificates. Reapplying for hard copies can be time-consuming because certificates are granted by different organizations and in-person application may be necessary. By contrast, applying for an e-copy can save paper and time. By providing information for identity verification, graduates are able to apply for any certificate easily. Nevertheless, because of this convenience, forged degree certificates,

licenses, and certificates are prevalent. Consequently, schools and companies cannot instantly validate the documents they receive. To solve this problem, a certificate system based on blockchain was designed in this study. Data are stored in different nodes, and anyone who wishes to modify a particular internal datum must request that other nodes modify it simultaneously. Thus, the system is highly reliable.

In this study, we developed a decentralized application and designed a certificate system based on Ethereum blockchain. This technology was selected because it is incorruptible, encrypted, and trackable and permits data synchronization. By integrating the features of blockchain, the system improves the efficiency operations at each stage. The system saves on paper, cuts management costs, prevents document forgery, and provides accurate and reliable information on digital certificates.



Fig: Blockchain Technology

1.2 Objectives

- To build a decentralized platform for issuing and verifying the genuinity of e-certificate.
- Other authorities like Company, University can verify the student details with pass status and mark.

1.3 System Specifications

- Hardware Requirements:-
  - Windows PC
- Software Requirements: -
  - Block Chain Tools
    - ✓ Ethereum Wallet (blockchain-based distributed computing platform)
    - ✓ Ganache BlockChain (Truffle)
    - ✓ Metamask
    - ✓ Etherium (Platform)
    - ✓ Solidity (Language)
- Application Requriments
  - Front End : HTML5, CSS3, BOOTSTRAP
  - Back end: JavaScript, PHP, MYSQL

CHAPTER 2
LITERATURE SURVEY

2.1 Blockchain

The concept of blockchain was proposed by Satoshi Nakamoto in 2008. Blockchain is an online ledger that provides decentralized and transparent data sharing. With distributed recordings, all transaction data (stored in nodes) are compressed and added to different blocks. Data of various types are distributed in distinct blocks, enabling verifications to be made without the use of intermediaries. All the nodes then form a blockchain with timestamps. The data stored in each block can be verified simultaneously and become inalterable once entered. The whole process is open to the public, transparent, and secure.

The emergence of Ethereum Smart Contracts in 2013 boosted blockchain technology, which became blockchain 2.0. As presented in Fig.1, blockchain 1.0 was mainly adopted by Bitcoin to solve problems concerning cryptocurrencies and decentralized payments. Blockchain 2.0 focused on decentralizing the entire market and is employed to transform assets through smart contracts, thereby creating value through the emergence of alternatives to Bitcoin .

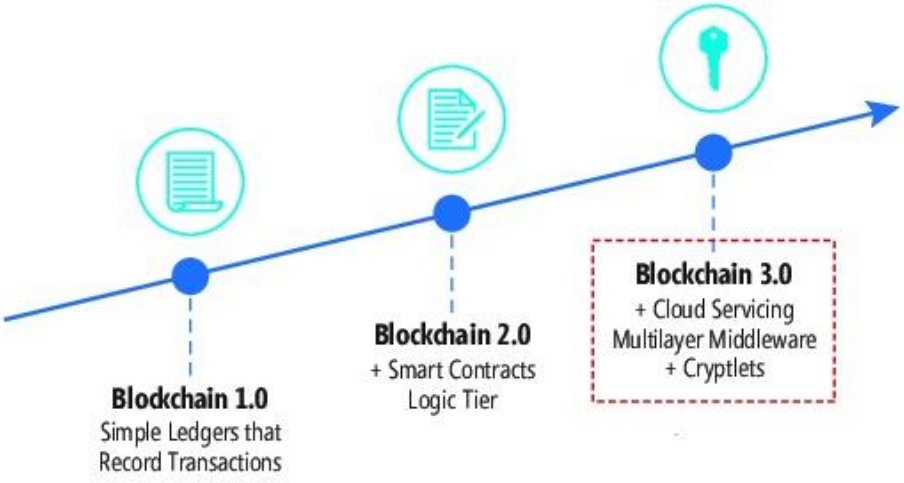


Fig.1Blockchain Technology Version

2.2 Ethereum

Ethereum is an open and decentralized platform featuring Turing completeness and supporting various derivative applications. Most smart contracts and decentralized autonomous organizations are created by using Ethereum. If the Bitcoin blockchains are considered a global payment network, Ethereum would be the global computing system. Furthermore, Ethereum is an open-source platform similar to Android (developed by Google). It provides an infrastructure that enables developers to create applications. The infrastructure is developed and maintained by both Ethereum and those developers. The major characteristics of Ethereum are as follows:

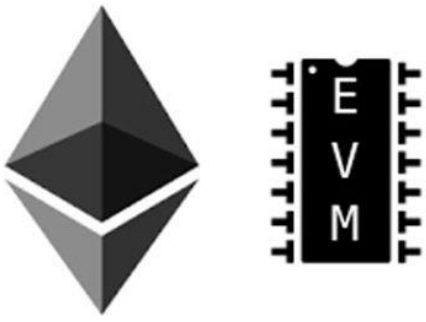
1) Incorruptible: third-parties are not able to modify any data;

2) Secure: errors derived from personnel factors are avoided because the decentralized applications are maintained by entities rather than individuals;

3) Permanent: blockchain does not cease to operate even if an individual computer or server crashes.

1) Ethereum Virtual Machine (EVM): The EVM is a programmable blockchain. Unlike Bitcoin, which provides a fixed set of commands, the EVM allows developers to run any programs in the manner they wish. Developers instruct the EVM to execute applications by using a high-level language called Solidity.

2) Solidity: Solidity is the programing language used for implementing smart contracts and is similar to JavaScript. After a Solidity-programmed smart contract is completed, a complier called solc is required to transform the Solidity code into contract bytecode, which is then interpreted by the EVM. Next, the compiled instructions are deployed in an Ethereum blockchain.



2.3 Ethereum Virtual Machine

Ethereum is a programmable blockchain. Rather than give users a set of pre-defined operations (e.g. bitcoin transactions), Ethereum allows users to create their own operations of any complexity they wish. In this way, it serves as a platform for many different types of decentralized blockchain applications, including but not limited to cryptocurrencies.

Ethereum in the narrow sense refers to a suite of protocols that define a platform for decentralized applications. At the heart of it is the Ethereum Virtual Machine ("EVM"), which can execute code of arbitrary algorithmic complexity. In computer science terms, Ethereum is "Turing complete". Developers can create applications that run on the EVM using friendly programming languages modelled on existing languages like JavaScript and Python.

The Ethereum platform itself is featureless or value-agnostic. Similar to programming languages, it is up to entrepreneurs and developers to decide what it should be used for. However, it is clear that certain application types benefit more than others from Ethereum's capabilities. Specifically, ethereum is suited for applications that automate direct interaction between peers or facilitate coordinated group action across a network. For instance, applications for coordinating peer-to-peer marketplaces, or the automation of complex financial contracts. Bitcoin allows for individuals to exchange cash without involving any middlemen like financial institutions, banks, or governments. Ethereum's impact may be more far-reaching. In theory, financial interactions or exchanges of any complexity could be carried out automatically and reliably using code running on Ethereum. Beyond financial applications, any environments where trust, security, and permanence are important – for instance, asset-registries, voting, governance, and the internet of things – could be massively impacted by the Ethereum platform.

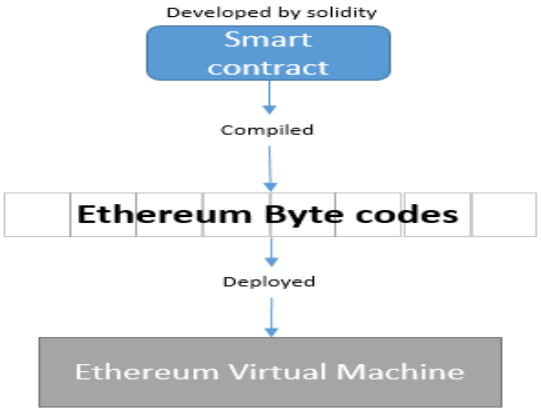


Fig.3 EVM

2.4 Smart contracts

Smart contracts were first proposed by Nick Szabo in the early 1990s. He explained that a smart contract enabled computers to execute transaction clauses. As blockchain has become popular, smart contracts have received increased attention. Smart contracts are the main feature of Ethereum, a blockchain platform founded in 2015. A smart contract is “a digital contract that is written in source code and executed by computers, which integrates the tamper-proof mechanism of blockchain”. Smart contracts can be created using the Ethereum blockchain.

Developers are able, according to their needs, to specify any instruction in smart contracts; develop various types of applications, including those that interact with other contracts; store data; and transfer Ethers. Additionally, smart contracts that are deployed in blockchain’s are copied to each node to prevent contract tampering. With related operations executed by computers and services provided by Ethereum, human error can be reduced to avoid disputes regarding such contracts.

Smart contracts are mostly used in voting system and cryptocurrency applications. Fig. 4 depicts an example of how developers can easily deploy smart contracts for cryptocurrency transactions. The high-level programming languages used for writing smart contracts are mainly Solidity, Serpent, and LLL . Currently, most developers employ Solidity to write smart contracts and compile the instructions into bytecode for the EVM to execute. Certain costs are incurred when developers create smart con



Fig.4 Smart Contract

2.5 Solidity

Solidity is a statically-typed programming language designed for developing smart contracts that run on the EVM. Solidity is compiled to bytecode that is executable on the EVM. With Solidity, developers are able to write applications that implement self-enforcing business logic embodied in smart contracts, leaving a non repudiable and authoritative record of transactions. Writing smart contracts in smart contract specific languages such as Solidity is referred to as easy (ostensibly for those who already have programming skills).

Compared to other EVM-targeting languages of the time such as Serpent and Mutan, Solidity contained a number of important differences. Complex member variables for contracts including arbitrarily hierarchical mappings and structs were supported. Contracts support inheritance, including multiple inheritance with C3 linearization.

CHAPTER 3
RESEARCH METHODS

3.1 System Design

In this study, a blockchain certificate system was developed based on relevant technology. The system's application was programmed on the Ethereum platform and is run by the EVM. In the system, three groups of users are involved, (Fig. 5). Schools or certification units grant certificates, have access to the system, and can browse the system database. When students fulfilled certain requirements, the authorities grant a certificate through the system. After the students have received their certificate, they are able to inquire about any certificate they have gained. The service provider is responsible for system maintenance.

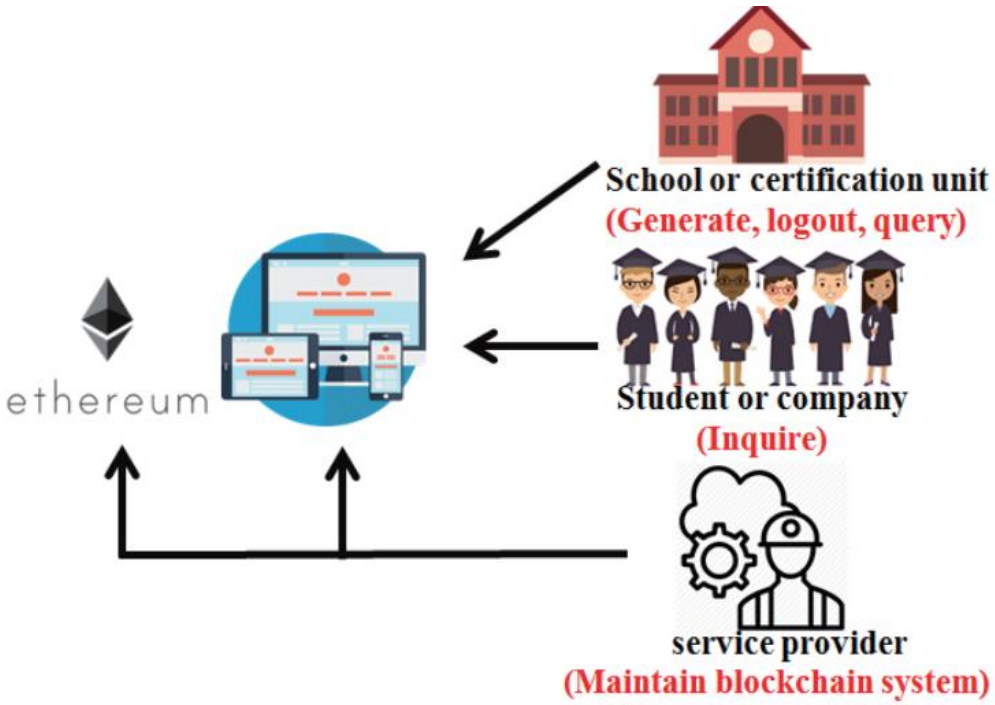


Fig.5 System Design

3.2 Process

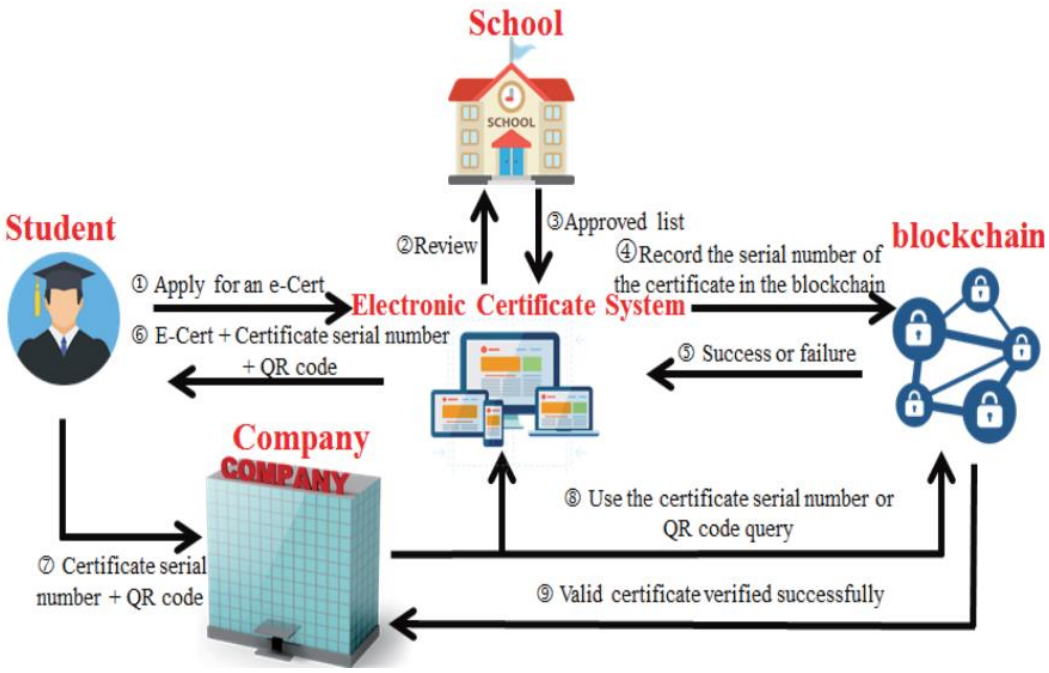


Fig.6 Process

Blockchain is a decentralized distributed database. The working processes of the system developed in this study are as follows:

1. Schools grant a degree certificate and enter the student's data into the system. Next, the system automatically records the serial number of the student in a blockchain.
2. The certificate system verifies all the data.
3. Instead of sending conventional hard copies, schools grant e-certificates containing a quick response (QR) code to the graduates whose data have been successfully verified. Each graduate also receives an inquiry number and electronic file of their certificate.
4. When applying for a job, a graduate simply sends the serial number or e-certificate with a QR code to the target companies.
5. The companies send inquiries to the system and are informed if the serial numbers are validated. The QR code enables them to recognize if the certificate has been tampered with or forged.

CHAPTER 4
PROPOSED SYSTEM

4.1 Module Description:

A web application with blockchain for issuing and validation of the certificates. The system will allow generating and verification of e - certificates issued by university/schools to students. Any entity can verify the genuineness of the certificate. The genuineness is ensured through using a blockchain.

4.2 System Features:

Reduces the workload of universities /schools in issuing an e-certificate. Students can easily save, share or print the e-certificate. They can also share the certificate id to the entity for verification. The smart contract deployed on the network ensure no errors happen during the store and retrieve process

System Modules:

- **COLLEGE/ UNIVERSITY**
  - Login
  - Requester list
  - Accept request
  - Student registration

- **STUDENT**
  - Login
  - Certificate request
  - View profile
  - View certificate

- OTHERS
  - Certificate verification

4.3 MODULES:

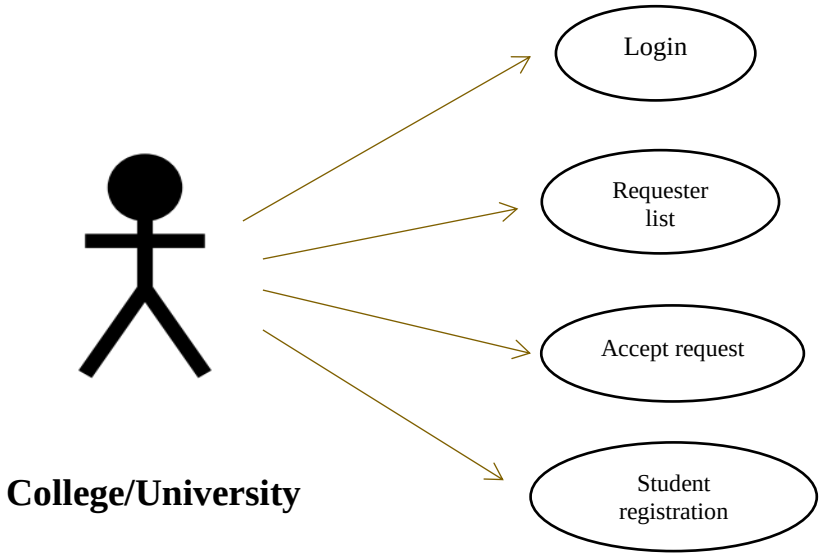
- **COLLEGE/UNIVERISTY**
  - Add student
    - College/University can only create login id for student and add student details to the database.
  - Issue Certificate
    - College gives permission for student to generate certificate.
- **STUDENT**
  - View Certificate
    - If college has permitted the student for generating the certificate student can generate certificate
  - Password Reset
- **OTHERS**
  - Verify Certificate
    - Companies can verify the genuineness of the student certificate with student id.

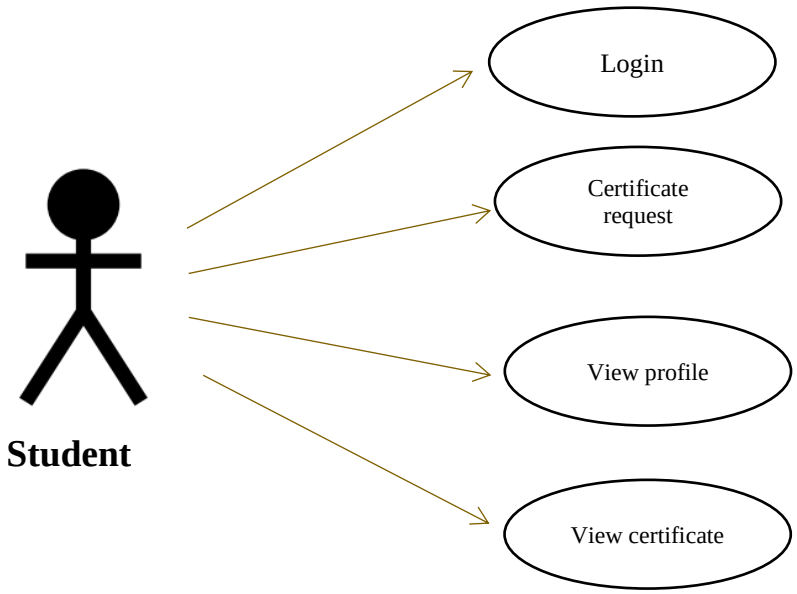
CHAPTER 5
DESIGN

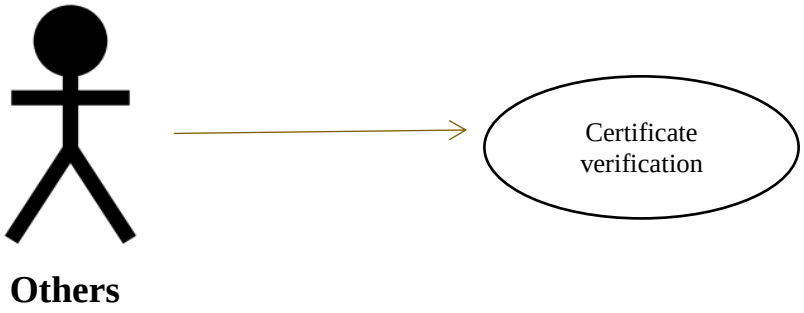
It is the first step in moving from problem domain to solution domain. The purpose of the design phase is to plan a solution of the problem specified by the requirements document. Starting with what is needed, design takes towards how to satisfy the needs.

The architecture defines the components, their interfaces and behaviors. The deliverable design document is the architecture. The design document describes a plan to implement the requirements. The system is designed to satisfy the requirements identified in the previous phases.

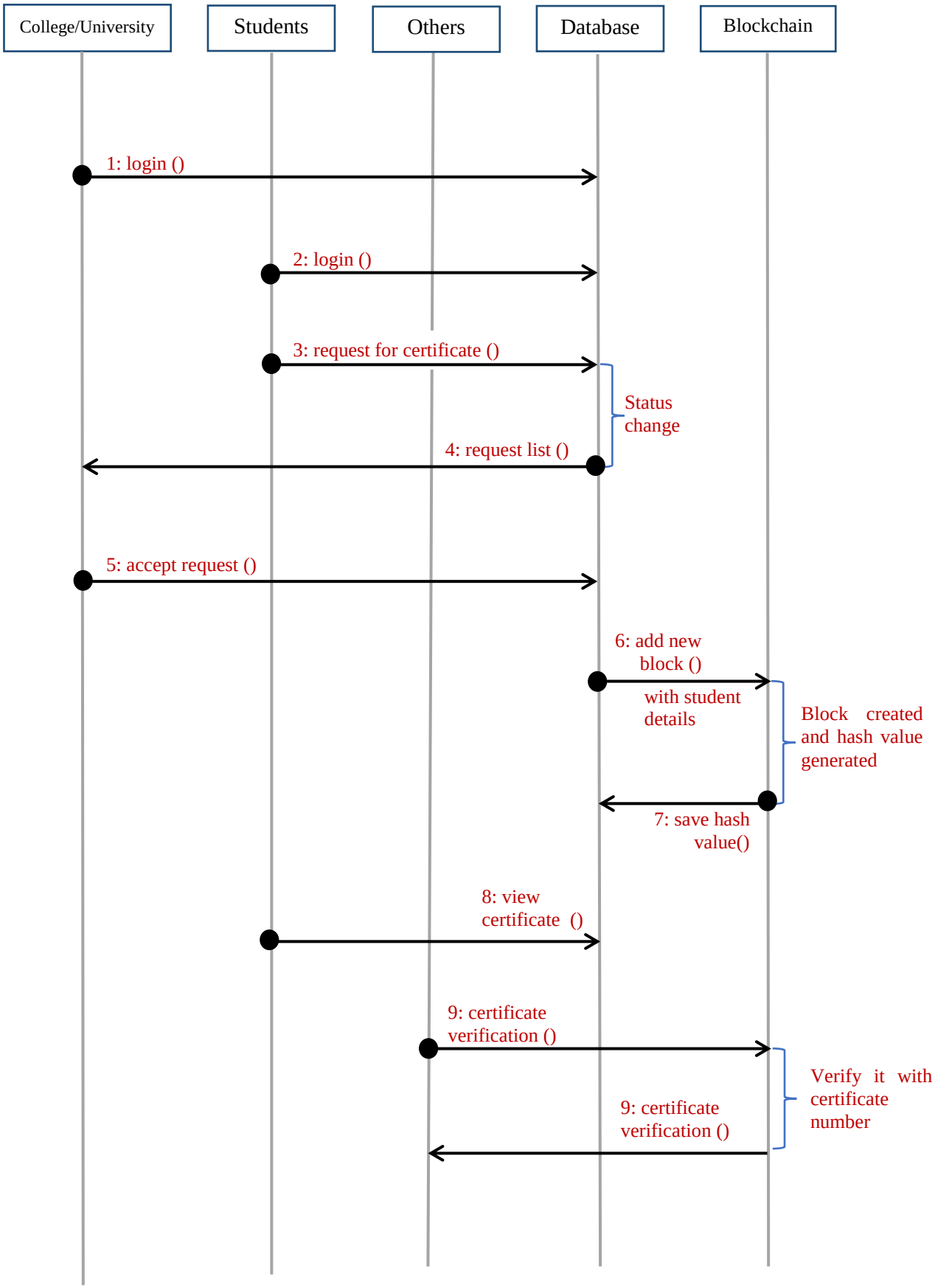
5.1 UML Diagrams:




Login
Certificate request
View profile
View certificate
Student


Certificate verification
Others

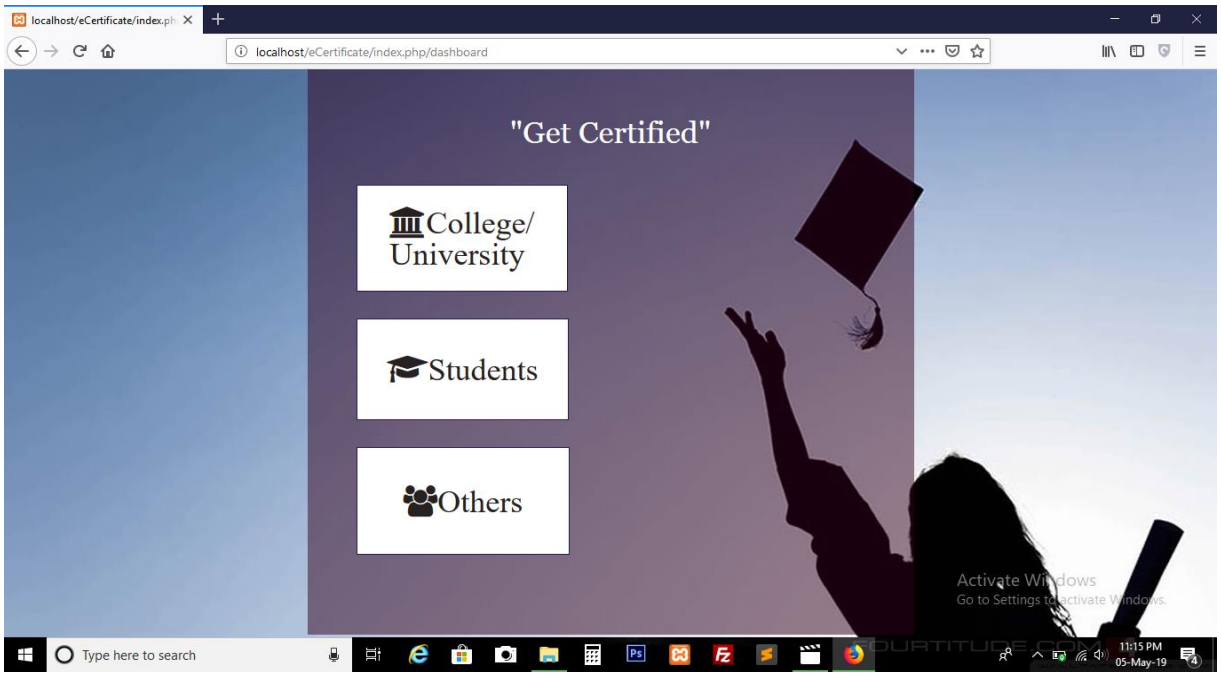
5.2 Sequence Diagram:



CHAPTER 6

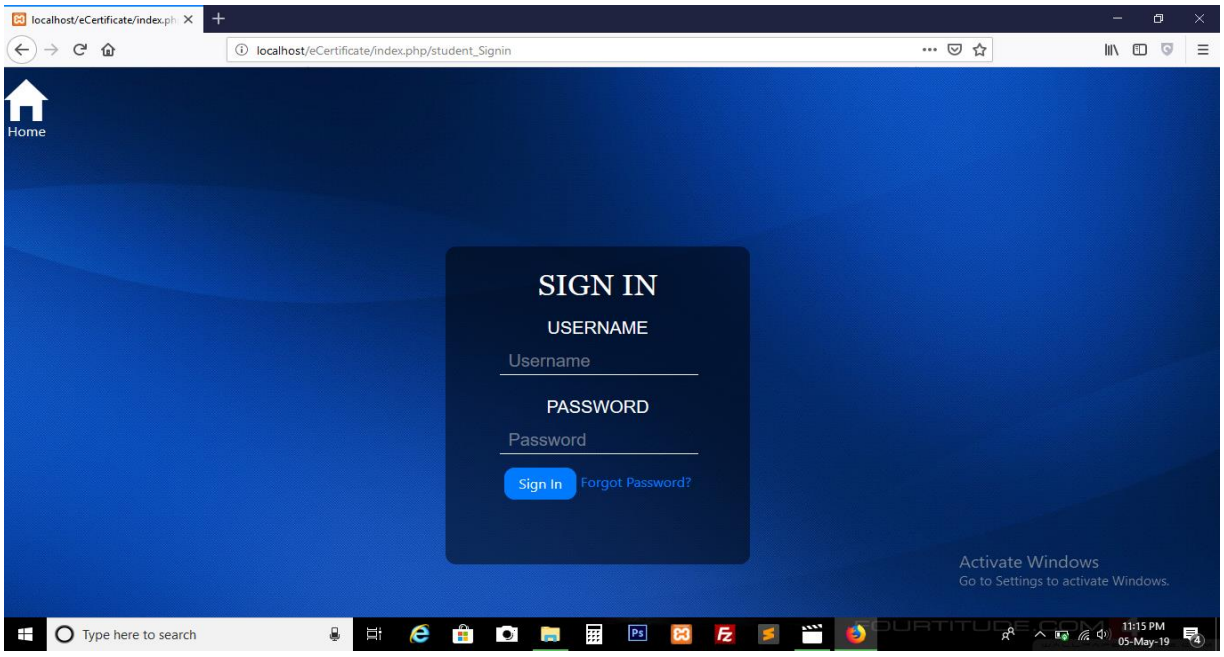
RESULTS

Home Page:



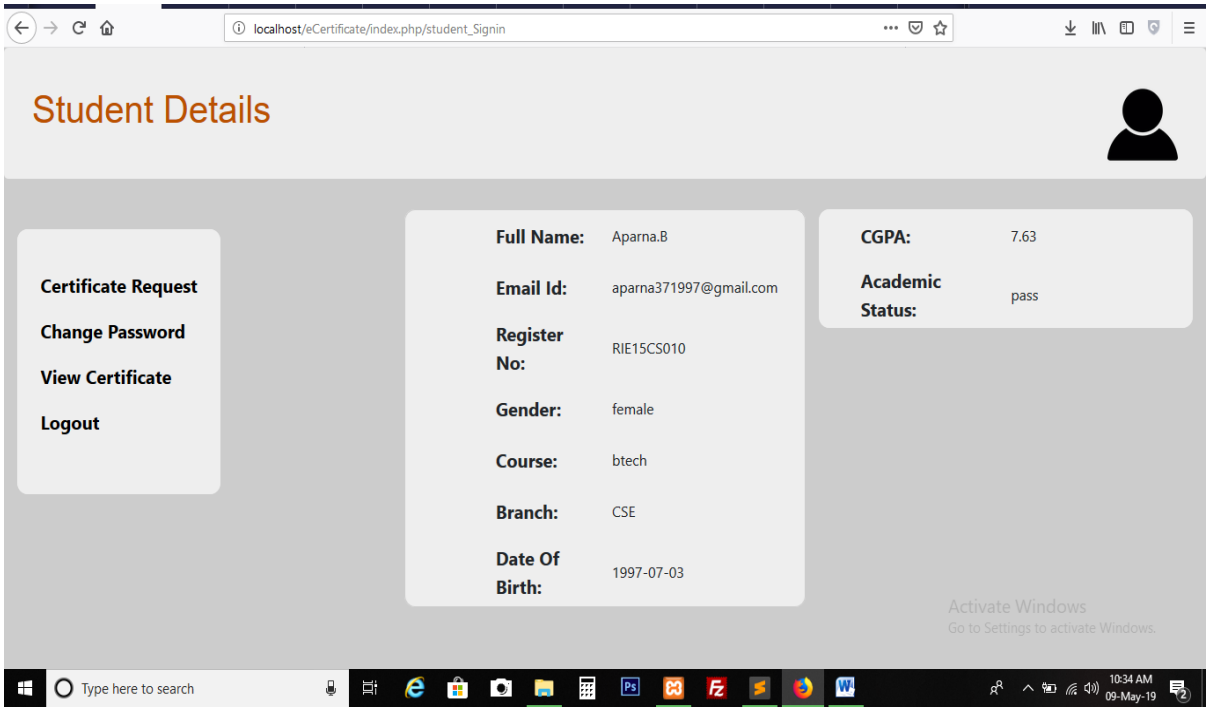
Home page with three modules college/university, Students and Others. Each modules enter into their respective page.

Student Login Page:



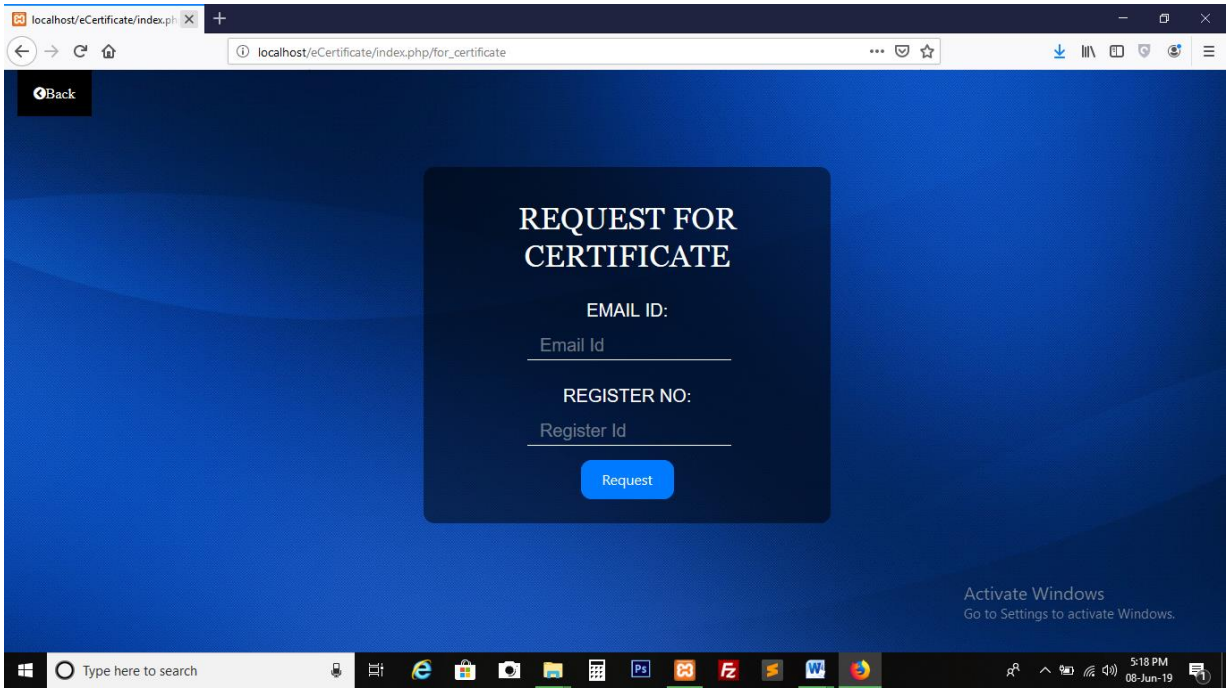
Student login page, login with their respective credential.

Student dashboard:

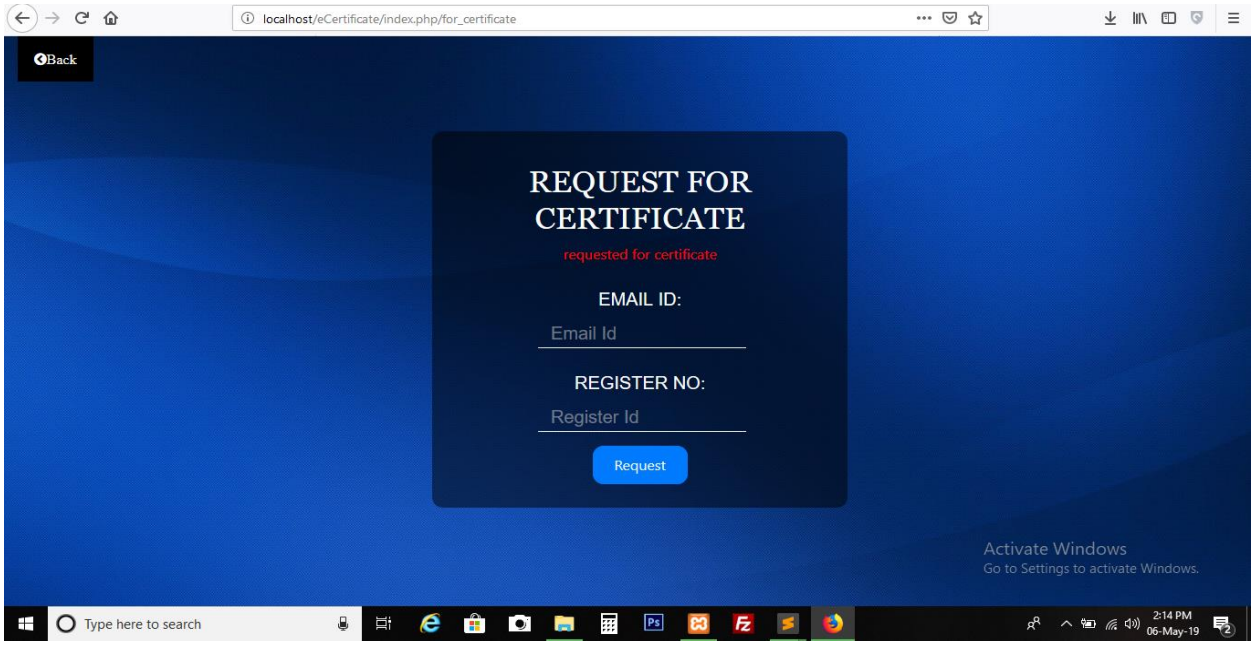


In student dashboard , Student profile is displayed and other options. Other options are Certificate request, Change Password, View Certificate and logout.

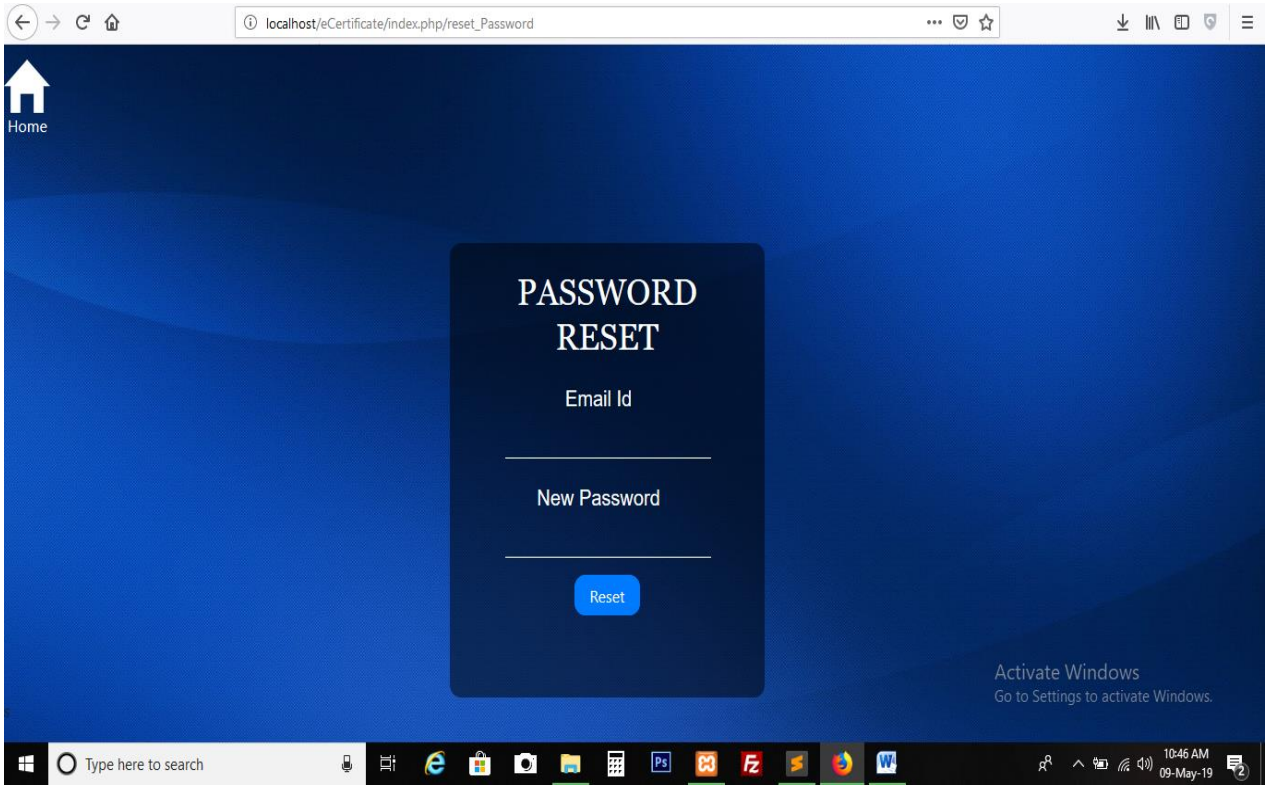
Certificate Request:



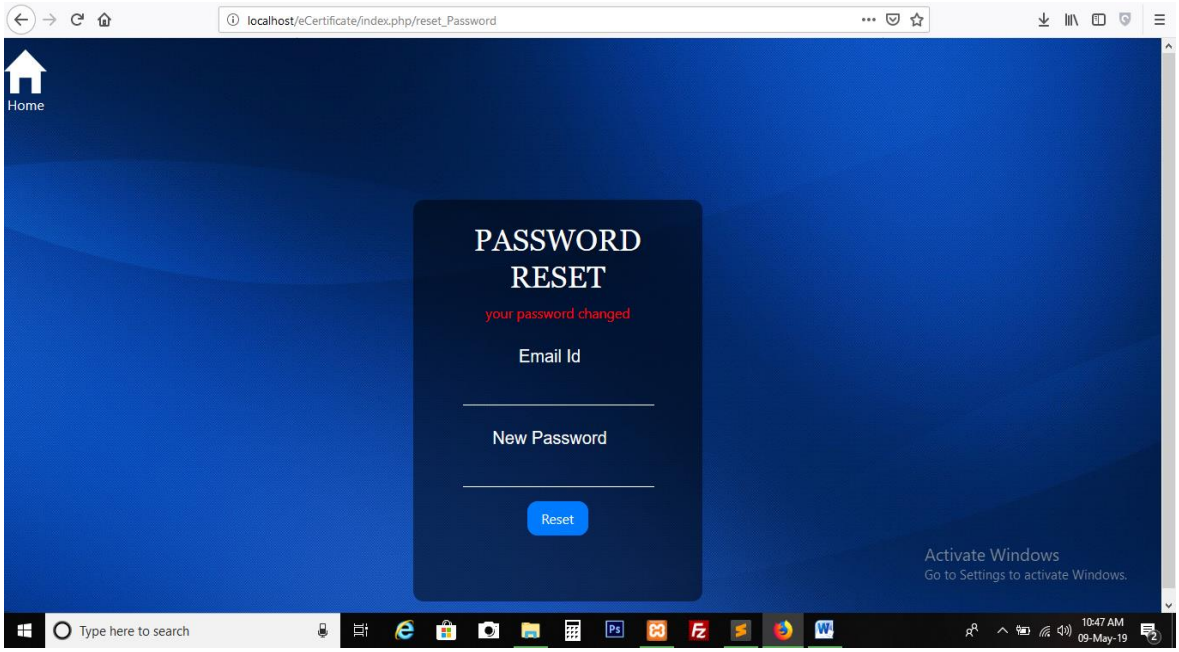
Certificate is requested with Email Id and Register Number. Once the request applied successfully a message displayed on the same page that “requested for certificate” as shown below.



Change Password:



Password can change by providing email id and new password then a message displayed that "your password changed" otherwise "invalid email id".

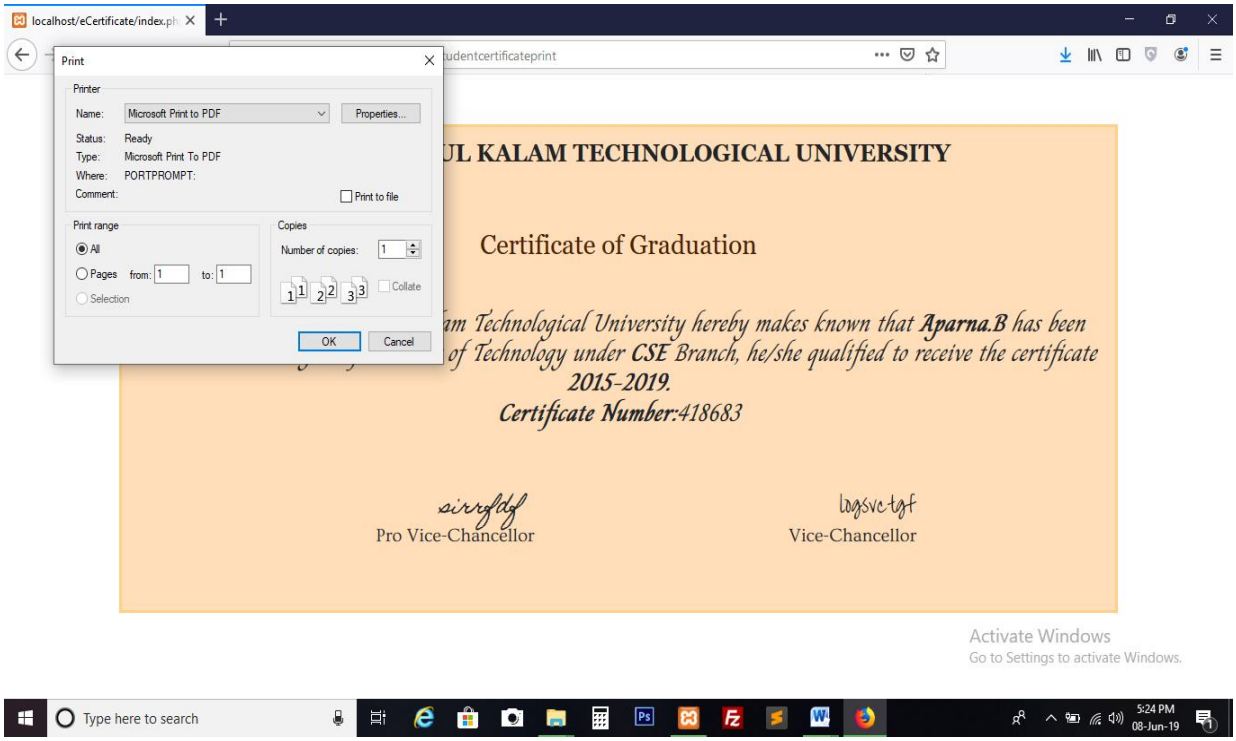


View Certificate:

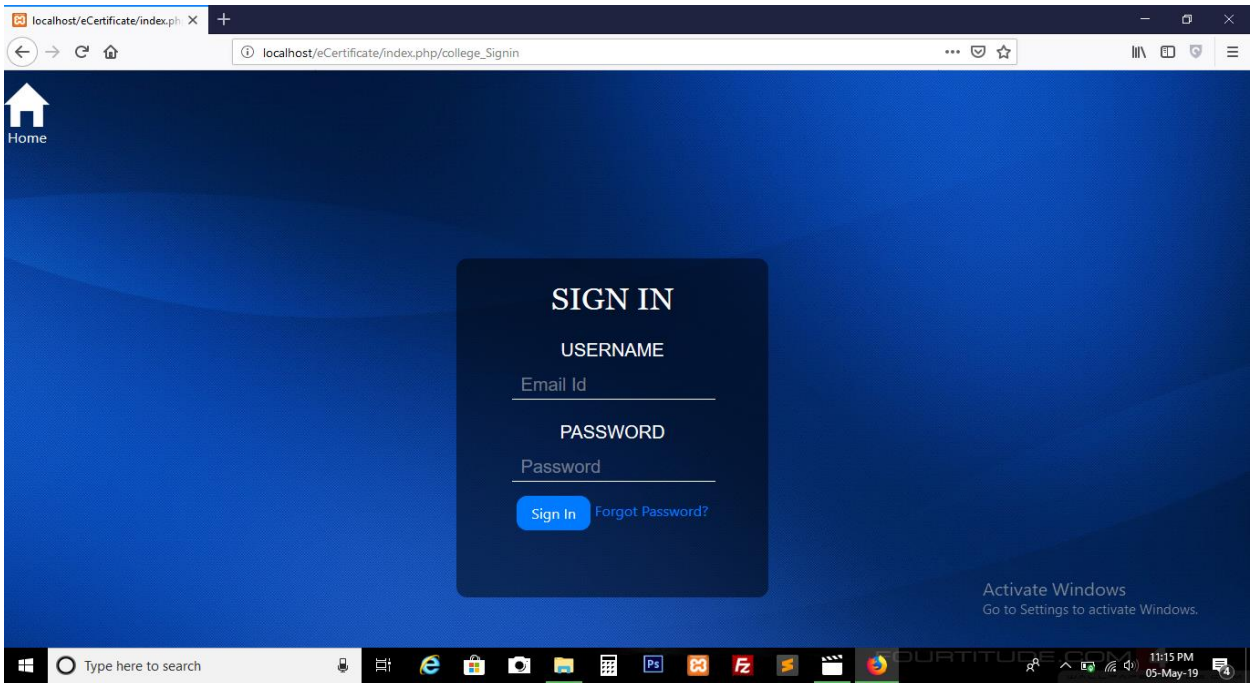
If a student didn't request for certificate, then certificate cannot be generated and the page to view certificate displays a message "request to generate your certificate", Otherwise the certificate displayed with a certificate number , name etc. as shown below.



There is a print button, the student print the certificate also.

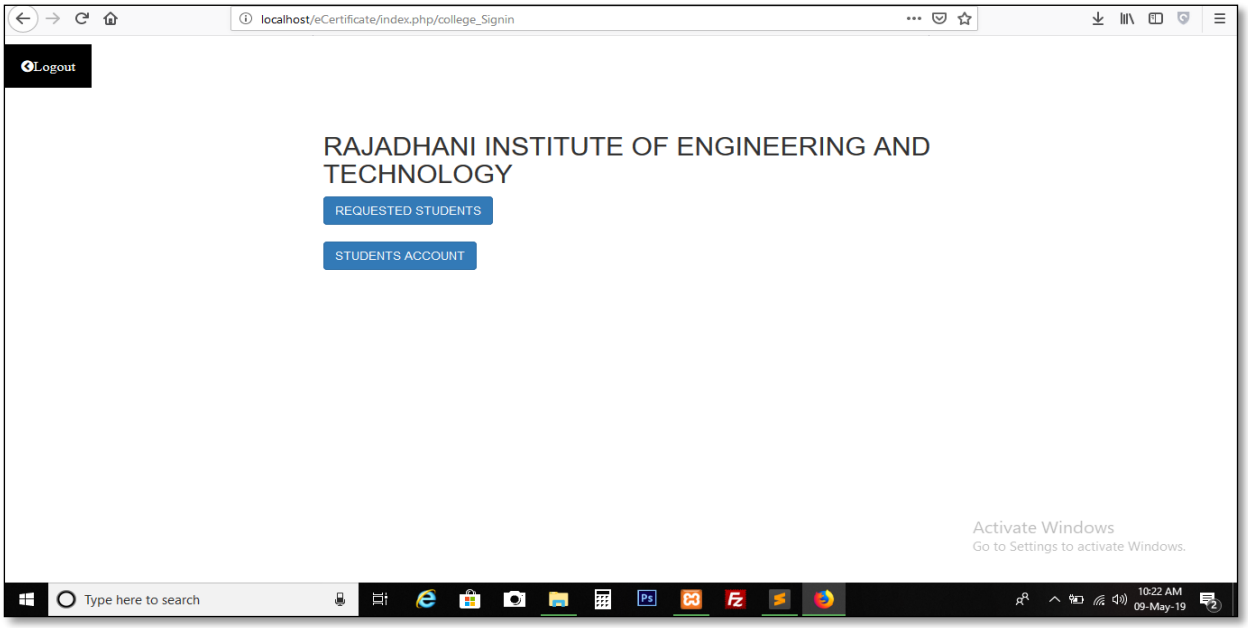


College Login:

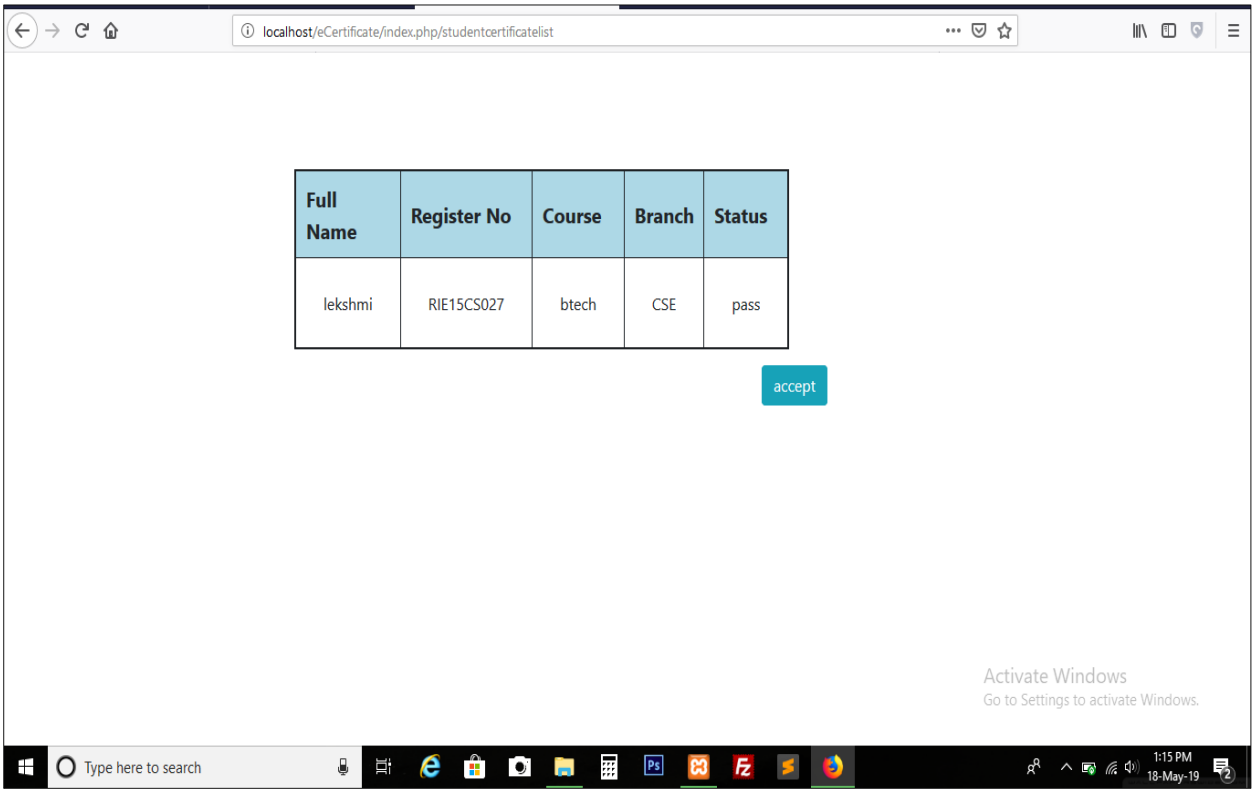


Student login page, login with their respective credential.

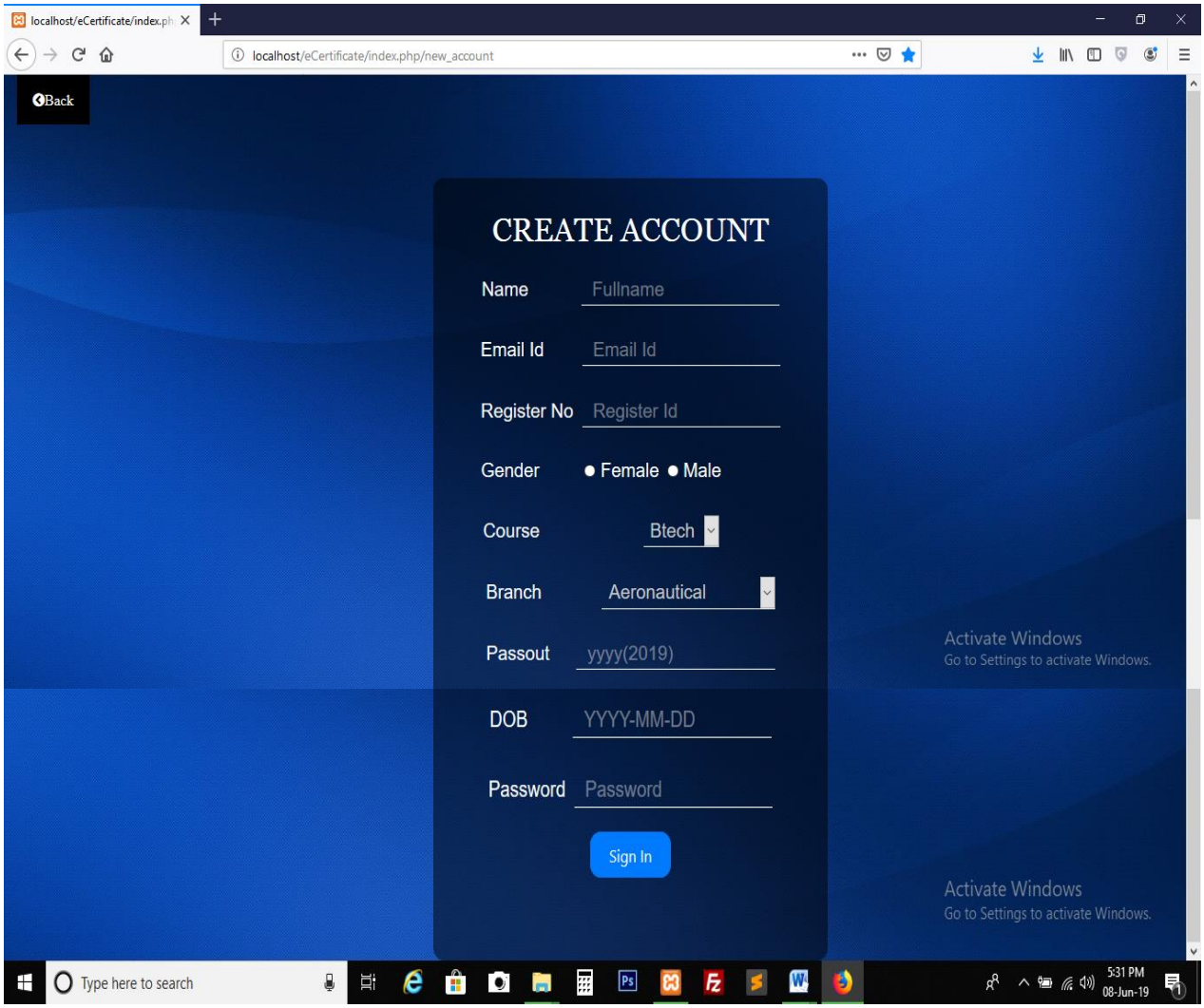
College Dashboard:



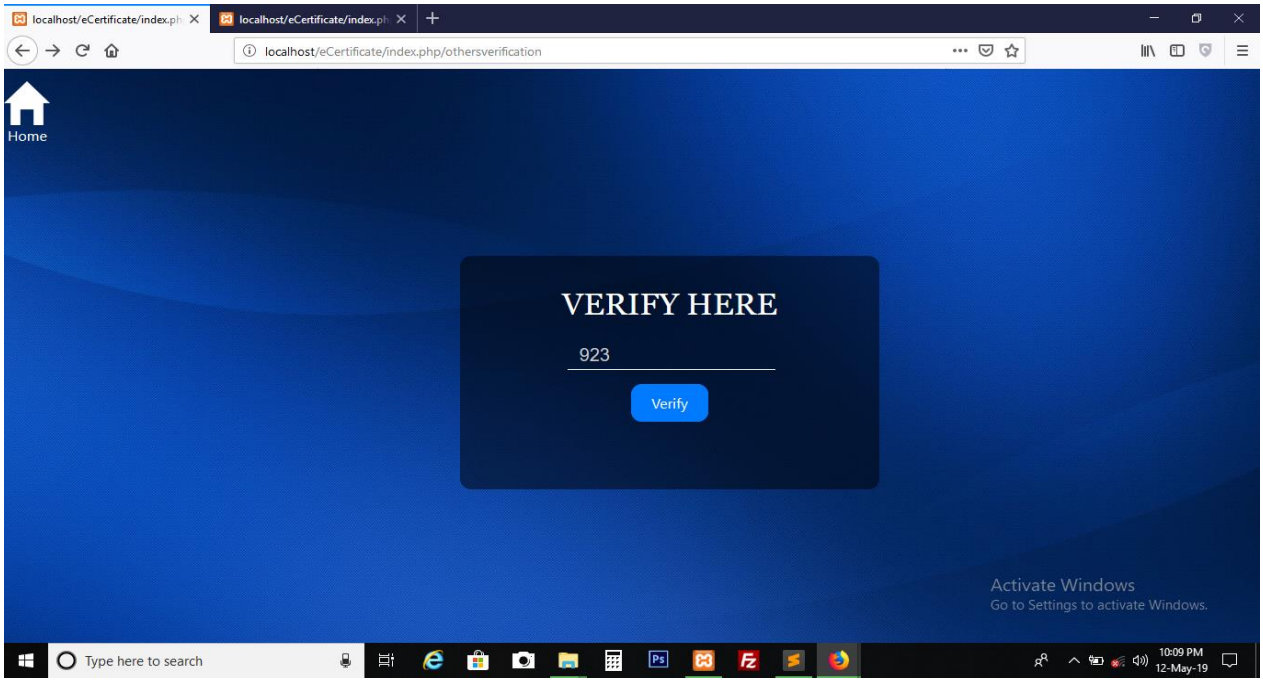
Requested Student List:



Students Account Creation:



Others:



Verify the student details by using their certificate number.

CHAPTER 7
SYSTEM STUDY

7.1 Feasibility Study

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the university. For feasibility analysis, some understanding of the major requirements for the system is essential.
Three key considerations involved in the feasibility analysis are

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

7.2 Non Functional Requirements

Non-functional requirements are the quality requirements that stipulate how well software does what it has to do. These are Quality attributes of any system; these can be seen at the execution of the system and they can also be the part of the system architecture.

7.3 Accuracy:

The system will be accurate and reliable based on the design architecture. If there is any problem in the accuracy then the system will provide alternative ways to solve the problem.

7.4 Usability:

The proposed system will be simple and easy to use by the users. The users will comfort in order to communicate with the system. The user will be provided with an easy interface of the system.

7.5 Accessibility:

The system will be accessible through internet and there should be no any known problem.

7.6 Performance:

The system performance will be at its best when performing the functionality of the system. performance testing is in general a testing practice performed to determine how a system performs in terms of responsiveness and stability under a particular workload.

7.7 Reliability:

Reliability of a system is defined as the probability that the product will not fail throughout a prescribed operating period .The proposed system will be reliable in all circumstances and if there is any problem that will be affectively handle in the design.

7.8 Security:

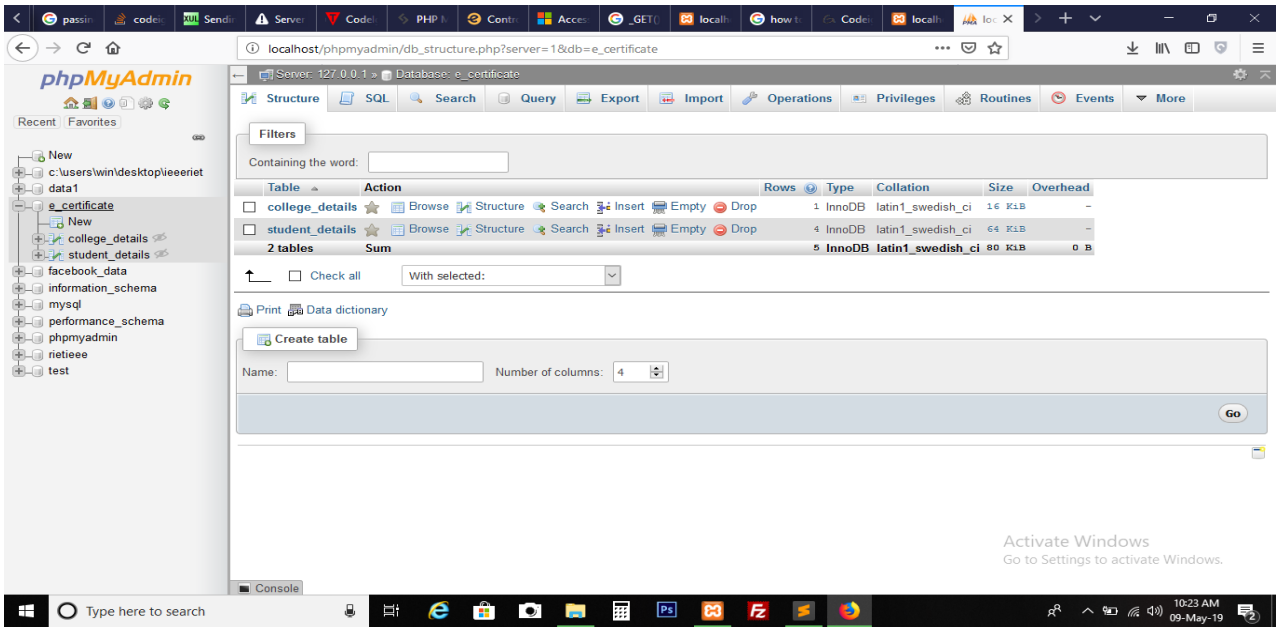
The proposed system will be highly secured; every user will be required registration and username/password to use the system. The system will do the proper authorization and authentication of the users based on their types and their requirements. The proposed system will be designed persistently to avoid any misuse of the application.

CHAPTER 8

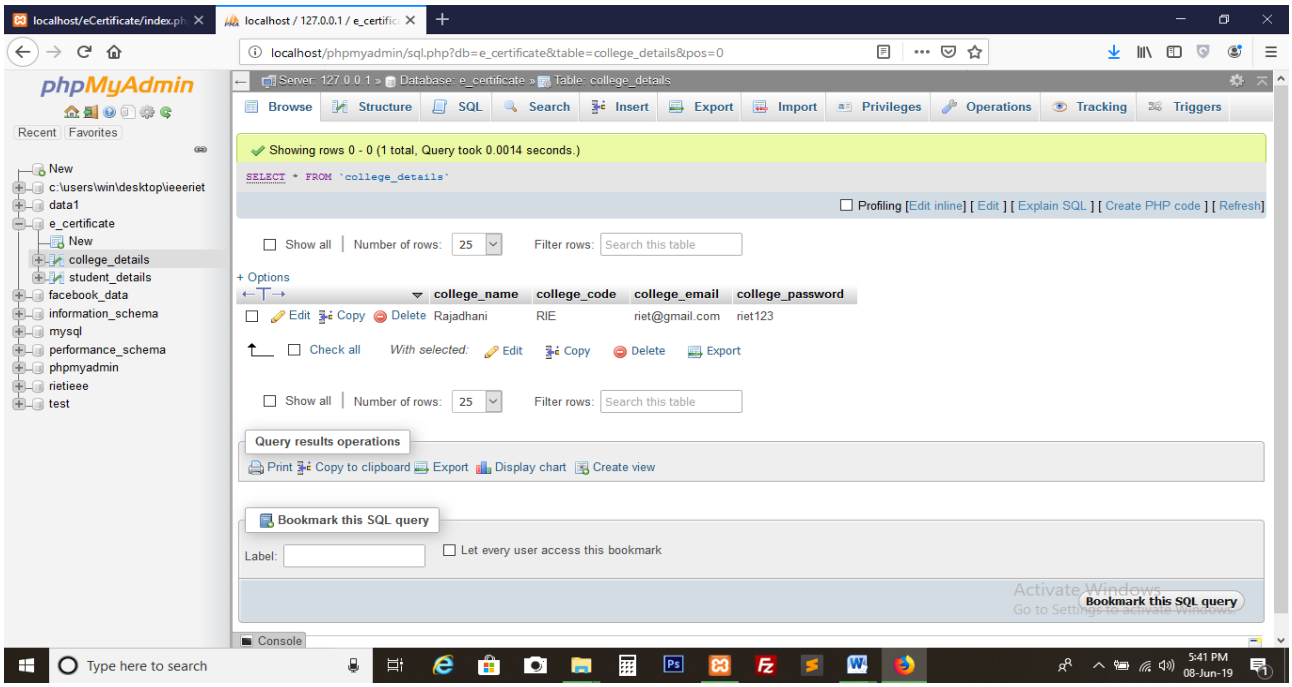
EVALUATION

Database:

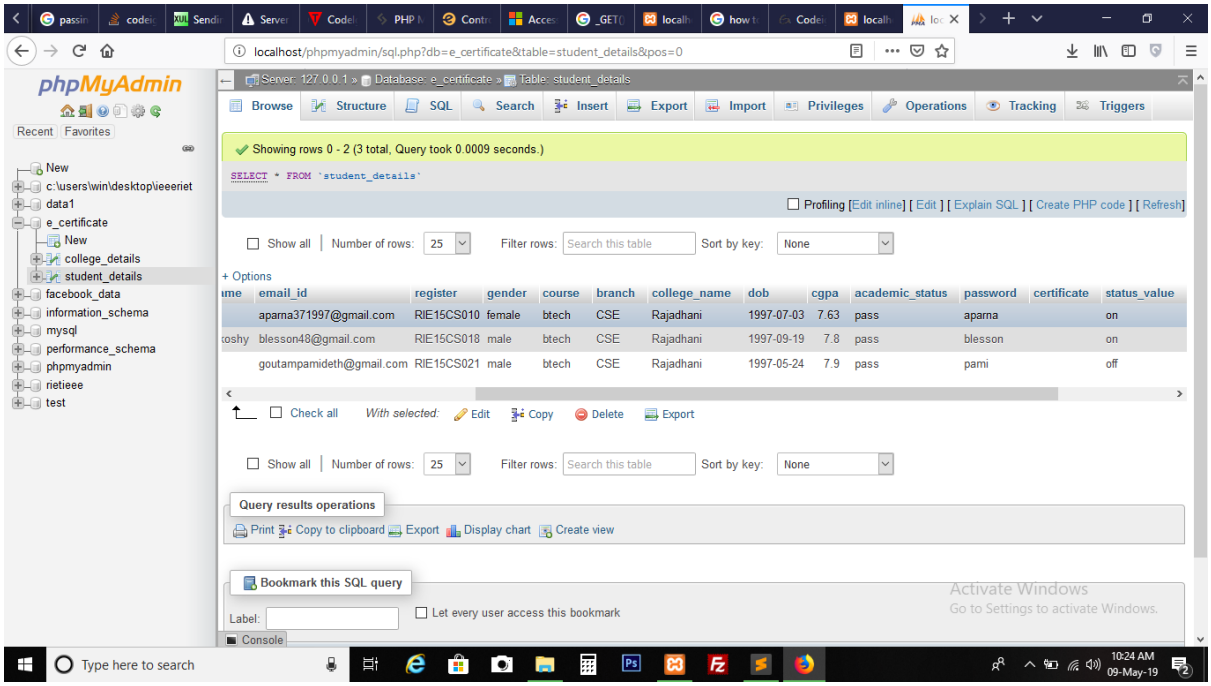
Two tables 1. For college/university. 2. For students.

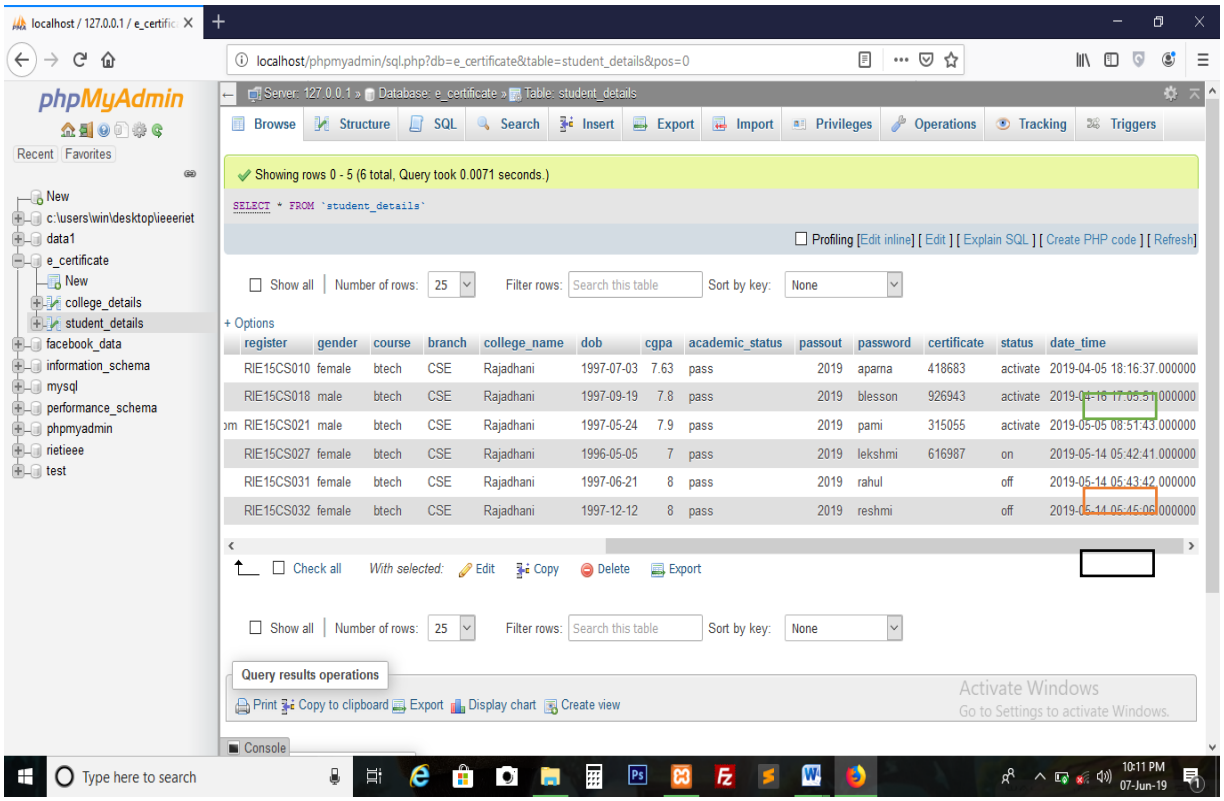


College_details table:



Student_details table:





off

If there is no student requested for the certificate then the status field is “off”-default value.

on

When the student request for certificate then the status field update to “on” then only the list is displayed in the college page.

activate

When the college authority accept the certificate request then the status field updated to “activate”.

CHAPTER 9
CONCLUSION

Data security is one of the major features of blockchain technology. Using the proposed blockchain based system reduces the likelihood of certificate forgery. The process of certificate application and automated certificate granting are open and transparent in the system. Companies or organizations can thus inquire for information on any certificate from the system. In conclusion, the system assures information accuracy and security.

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